# Retail Inventory & Sales Analytics

## End-to-end portfolio project | MySQL, Python, Power BI and Excel

# Executive summary

This project demonstrates an end-to-end retail analytics workflow using synthetic 2025 sales transactions, product and store dimensions, and a year-end inventory snapshot. It is designed to help a retailer understand revenue, profitability, demand patterns and inventory risk.

# Business objectives

• Measure sales and gross profit trends.

• Compare category, product and store performance.

• Flag stock at or below reorder levels.

• Identify slow-moving and potentially overstocked items.

• Recommend data-driven replenishment and inventory actions.

# Dataset overview

| Dataset | Grain | Rows |
|---|---|---|
| Products | One row per product | 50 |
| Stores | One row per store | 6 |
| Sales | One product line per order | 25,370 |
| Inventory | One product-store snapshot | 300 |

# KPI definitions

| KPI | Definition |
|---|---|
| Revenue | Sum of net sales amount after discounts |
| Gross Profit | Net sales amount minus cost amount |
| Gross Margin % | Gross profit divided by revenue |
| Orders | Distinct order IDs; do not count sales lines |
| Average Order Value | Revenue divided by distinct orders |
| Inventory Cost Value | On-hand stock multiplied by unit cost |
| Low Stock | Stock quantity less than or equal to reorder level |
| Months of Cover | On-hand stock divided by recent average monthly units sold |

# Business recommendations framework

• Low stock: prioritize replenishment where on-hand stock is below reorder level and recent sales velocity is high.

• Overstock: review purchase plans, inter-store transfers, bundles and markdowns for items with excessive months of cover.

• Slow-moving products: investigate pricing, assortment relevance and product visibility before reordering.

• High-performing products: protect availability and consider demand-based safety stock.

• Store performance: compare revenue and average order value while accounting for local demand and store size.

# Limitations and next steps

The dataset is synthetic and intended for learning. Inventory is a single 31-Dec-2025 snapshot, so a rigorous inventory turnover ratio cannot be calculated. Add monthly inventory snapshots, purchase orders, receipts, returns, stock adjustments and stockout events for a production-grade model.

# Suggested portfolio presentation

Show the data model, three or four key SQL queries, a Python EDA notebook, screenshots of the Power BI pages, and three evidence-based business recommendations. Clearly label the dataset as synthetic and replace illustrative statements with measured results from your analysis.